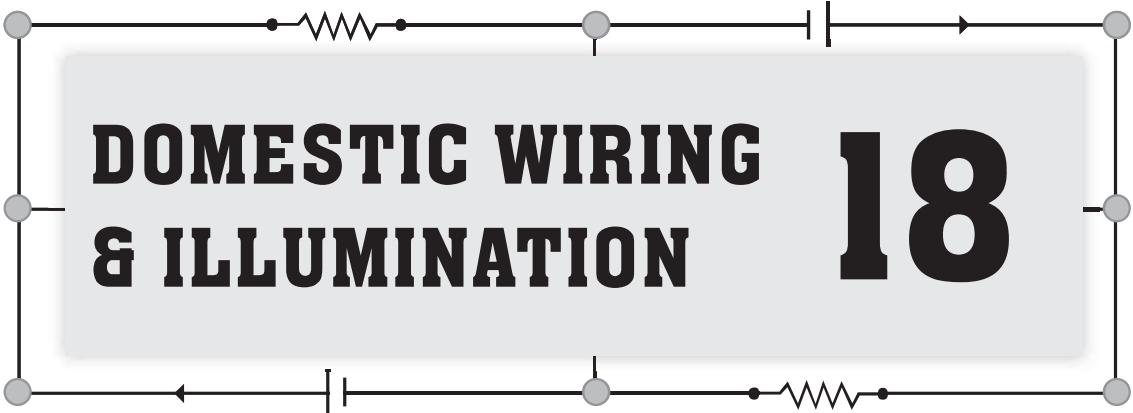




DOMESTIC WIRING & ILLUMINATION

18

LEARNING OBJECTIVES

After the completion of this chapter, the students or readers will be able to understand the following:

- ❖ What are different types of cables used in wiring installations?
- ❖ Which are different types of wiring systems used in domestic, commercial, and industrial installations?
- ❖ How different wiring circuits are designed?
- ❖ What is illumination and what are different terms used their in?
- ❖ What are various laws of illumination?
- ❖ What are different types of electric lamps and what are their characteristics?
- ❖ What are different factors on which illumination depends?
- ❖ How lighting schemes are designed for different places as per required illumination

18.1 INTRODUCTION

The electric supply is given by the supplier up to energy meter. After energy meter, the electric supply is conveyed to the different places of the building with the help of a wiring system. Therefore, the wiring should be economical, safe, and good looking, which is the chief requirement of a building. With the modern developments, wiring techniques have undergone radical changes. In this chapter, various types of wiring systems and their applications are discussed.

18.2 TYPES OF CABLES

It should be necessary to know about various types of cables (insulated conductors) that are used for internal wiring systems before considering the various types of wiring systems suitable for any installation.

A solid or stranded conductor covered with insulation is known as a cable.

The cable may be a single core or multicore depending upon the number of conductors. Various types of insulating materials are employed for covering the conductors.

Accordingly, the cables (wiring conductors) may be classified as follows:

1. Vulcanized Indian Rubber (VIR) cables
2. Polyvinyl chloride (PVC) cables
3. Tough rubber sheathed (TRS) or Cab tire sheathed (CTS) cables
4. Lead sheathed cables
5. Weather proof cables

18.3 TYPES OF WIRING SYSTEMS

The main types of wirings usually employed in residential buildings, commercial buildings, and industries are as follows:

1. Cleat wiring
2. Casing and capping wiring
3. CTS or TRS wiring
4. Metal-sheathed wiring
5. Conduit wiring

18.3.1 Cleat Wiring

In this system of wiring, usually VIR or PVC conductors are employed. The conductors are supported in porcelain cleats that are placed at least 6 mm above the walls. The porcelain cleats are made of two parts, namely base and cap, that is, the lower one is known as base being having two or three grooves for the accommodation of conductors and the upper one is known as cap as shown in Figure 18.1. The conductors are run in the grooves, cap is placed over the base, and the whole assembly is fixed on to the wall with the help of wooden screws and gutties (wooden or PVC plugs) already cemented in the wall. The screw not only fixes the cleats on the wall but also tightens the grip of the wires between the two halves of the cleat.

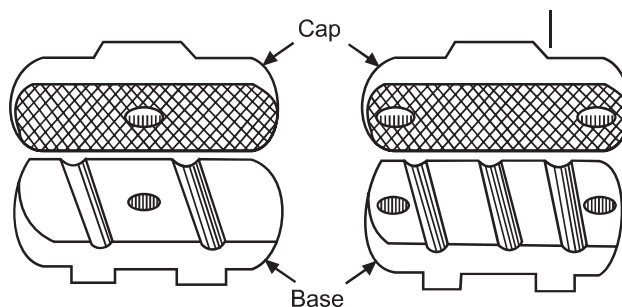


Fig. 18.1 Porcelain cleats

Advantages of cleat wiring

1. It is the cheapest system of wiring.
2. A little skill is required to lay the wiring.
3. This wiring can be installed very quickly.
4. It is the most suitable system for temporary wiring.
5. The wiring can be dismantled and recovered very quickly.
6. Inspection, alteration, and additions can be made easily.

Disadvantages of cleat wiring

1. It gives a rubbish look.
2. It is rarely employed for permanent job.
3. The insulation of wire is damaged while whitewashing or distempering the lime falls over the wires.
4. Oil and smoke are also injurious to VIR.
5. Mechanical injuries may damage the conductor since there is no protecting cover.

18.3.2 Casing and Capping Wiring

In this system of wiring, generally, VIR wires are employed. The casing is a base that consists of rectangular PVC or wooden block of seasoned teak wood and has usually two grooves to accommodate wires. The casing is fixed on the wall with the help of wooden screws and gutties already cemented in the wall. The casing is usually placed 3 mm apart from the wall by means of porcelain discs in order to protect the casing from dampness. Then, the wires of opposite polarity are laid in different grooves. After placing the wires in the grooves of casing, the casing is covered by means of a rectangular strip of seasoned wood of the same width as the casing known as capping with the help of wooden screws. The assembled view of casing and capping with the VIR wires placed in the grooves is shown in Figure 18.2.

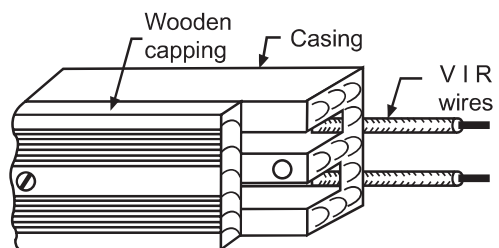


Fig. 18.2 Casing-capping wiring

Advantages of casing and capping wiring

1. It gives better appearance than cleat wiring.
2. Its cost is quite low as compared to other systems of wiring except cleat wiring.
3. It is easy to install and repair.
4. Conductors are strongly insulated.
5. Capping provides protection against mechanical injury.

Disadvantages of casing and capping wiring

1. It is not suitable in damp situations.
2. There is a risk of fire.
3. To make the job good looking highly skilled labour is required.

18.3.3 Cab Tire Sheathed or Tough Rubber Sheathed Wiring

In this system of wiring, generally CTS (Cab Tire Sheathed) or TRS (Tough Rubber Sheathed) conductors are employed. These conductors are run on well-seasoned, perfectly straight, and well-varnished teak wood batten of thickness 13 mm. The width of the batten is chosen depending upon the number of wires to be run on it. While doing this type of wiring, the batten is fixed on to the wall by means of wooden screws and gutties already cemented in the wall. The wires are held on the batten with the help of clips already fixed on the batten with the help of nails as shown in Figure 18.3.

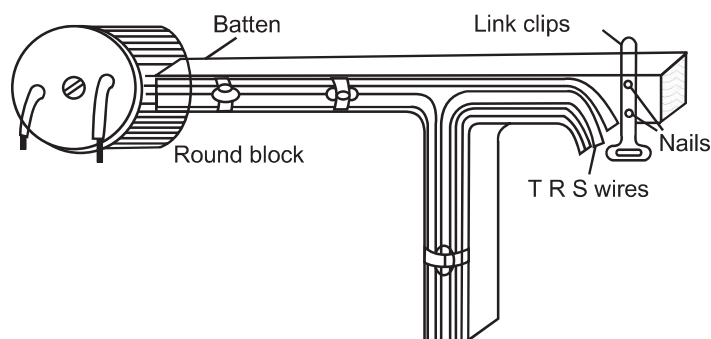


Fig. 18.3 CTS or TRS wiring

Advantages of tough rubber sheathed wiring

1. It is easy to install and repair.
2. It gives nice appearance.
3. The conductors have strong insulation, and therefore, it has longer life.
4. It is fireproof up to some extent.
5. Chemicals do not affect the conductor's insulation.

Disadvantages of tough rubber sheathed wiring

1. The conductors are open and liable to mechanical injury, and therefore, this type of wiring cannot be used in workshops.
2. Its use in places open to sun and rain is restricted.

18.3.4 Metal-sheathed Wiring

This system of wiring is similar to CTS or TRS wiring. The main difference is that, in this case, VIR conductors covered with lead alloy sheath (metal-sheathed cable) are used. The metal-sheathed cables are run on the wooden batten. The batten is fixed on the wall by means of screws and gutties already cemented into the wall. The cables are held on the batten with the help of link clips.

Advantages of metal-sheathed wiring

1. The conductors are protected against mechanical injury.
2. It can be suitably employed under damp situations.
3. It gives better appearance.
4. It has longer life.

5. Conductors are protected against chemicals.
6. It can be installed in open space.

Disadvantages of metal-sheathed wiring

1. The metal-sheathed cables are costlier than CTS or TRS wires.
2. In case of leakage, there is every risk of shock.
3. Skilled labour and proper supervision is required.

18.3.5 Conduit Wiring

In this system of wiring, VIR or TRS conductors are run in metallic or PVC tubes called conduits. This system of wiring provides best mechanical protection, safety against fire and shock. Therefore, it is considered to be the most suitable system of wiring for workshops and commercial buildings. The conduits can be either supported over the wall by means of saddles or buried under plaster.

Accordingly, there are two types of conduit wirings, namely surface conduit wiring and concealed conduit wiring. In surface conduit wiring, the conduit is run over the wall supported by means of saddles as shown in Figure 18.4, whereas in concealed conduit wiring, the conduit is embedded in the walls and ceiling by placing it in the cavity made previously in them. After placing the conduit, the insulated conductors (or cables) are drawn into them by means of G.I. (galvanized iron) wire known as pilot wire. A number of inspection boxes (conduit boxes) are provided along the run of conduit to facilitate the drawing of wires.

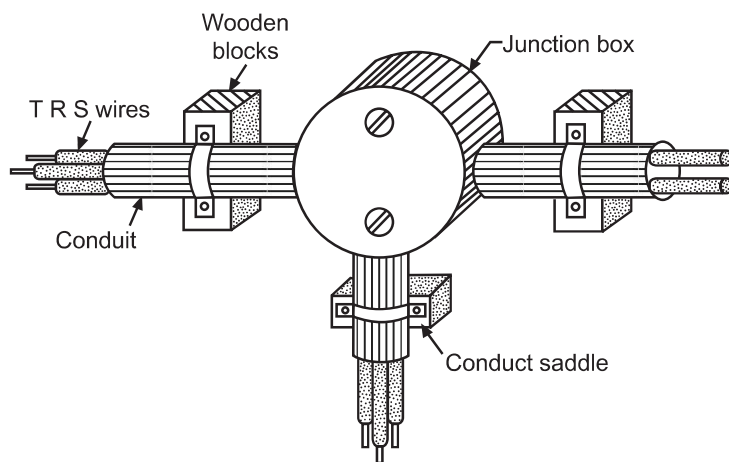


Fig. 18.4 Surface conduit wiring

Advantages of conduit wiring

1. Conduit provides protection against mechanical injury and fire.
2. Conduit provides protection against chemicals.
3. Conductors are safely secured from moisture.
4. This wiring has far better look.
5. It has a longer life.

Disadvantages of conduit wiring

1. It is costly system of wiring.
2. It requires more time for erection.
3. It requires highly skilled labour.

18.4 IMPORTANT LIGHTING ACCESSORIES

Some of the common lighting accessories are as follows:

1. **Main switch:** It is provided after the energy meter. It makes and breaks the phase and neutral connections simultaneously and controls the internal wiring circuits. In early days, switch-fuse units enclosed in iron boxes called iron clad switches were used. The following types of iron clad switches are employed in practice:
 - (a) **Double-pole iron clad (DPLC) switches:** These are used to control single-phase, two-wire circuits.
 - (b) **Triple-pole iron clad (TPIC) switches:** These are used to control three-phase, three-wire circuits.
 - (c) **Triple-pole iron clad switches with neutral link (TPNIC):** These are used to control three-phase, four-wire circuits.

However, nowadays, improved form of switches is employed. Fuse units with indicator light are used, which give better appearance.

2. **Switch:** Switch is a device that makes or breaks the circuit of an appliance such as lamp, fan, tube, socket, etc. There are two types of switches generally used in wiring system.
 - (a) **Surface or tumbler switch:** A tumbler switch is mounted on a round block or a teak wood board.
 - (b) **Flush switch:** A flush switch is generally mounted on teak wood board or Bakelite sheet.

Mostly, it is used in concealed conduit wiring system.

3. **Ceiling rose:** It is used to connect the pendent lamp holder, fan, or fluorescent tube to the wiring installation through flexible wire.
4. **Three-pin outlet socket:** It is used to give temporary connection of electrical appliances such as radios, table fans, television sets, electric iron, etc.
5. **Lamp holder:** It is used to hold the lamp. It has two spring-loaded pins to which phase and neutral is connected. The electric supply is conveyed through these pins to the lamp. There are various types of lamp holders (e.g., batten holder, pendent holder, angle holder, bracket holder, water-tight bracket holder, swivel lamp holder, etc.) employed in the wiring system depending upon the requirement.

18.5 IMPORTANT CIRCUITS

The supplier (P.S.E.B. Department) gives electric supply to energy meter. From the energy meter, the supply is connected to the consumer's main switch as shown in Figure 18.5, which controls the supply of internal wiring. From main switch, the supply is connected to main distribution board that carries a number of fuses and neutral link. Then, the supply is given to sub-main distribution board or to switch board.

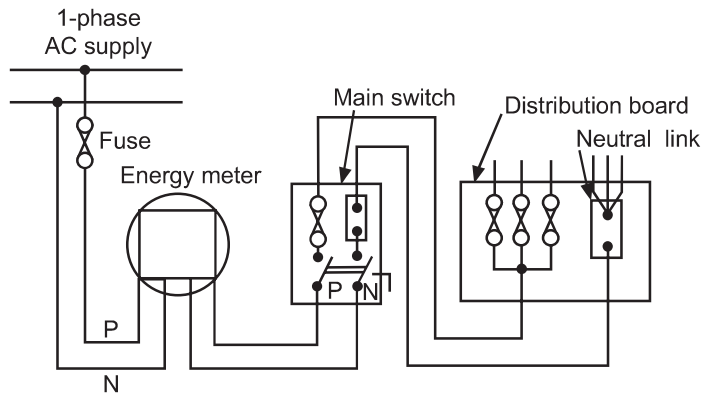


Fig. 18.5 Energymeter and distribution board

The important wiring circuits are as follows:

1. **Circuit to control one lamp with one switch:** The circuit is shown in Figure 18.6. In this figure, the switch is in the off position. The phase wire is connected to the lamp through switch, whereas the neutral wire is connected directly.

A switch is always connected in phase, so that when switch is off and some maintenance is done at the holder, the mechanic may not get shock.

2. **Circuit to control one lamp with two 2-way switches:** The circuit is shown in Figure 18.7. This circuit is used in staircase wiring. The two positions of the switches are shown. When both the switches are in the same position (i.e., either ON or OFF) as shown in Figure 18.7(a), the lamp will glow.

When the two switches are in different positions (i.e., one is in ON position and the other is in OFF position) as shown in Figure 18.7(b), the lamp will remain dark.

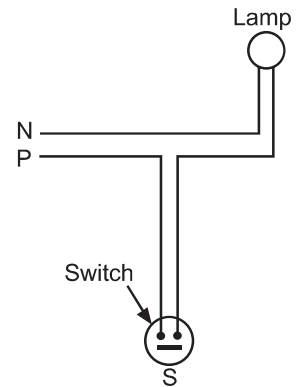


Fig. 18.6 One lamp controlled by one single-way switch

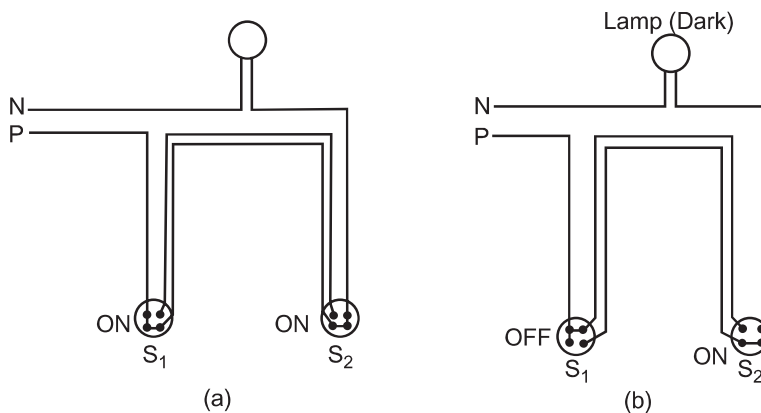


Fig. 18.7 One lamp controlled by two 2-way switches (a) Glowing position and (b) Dark position

3. **Circuit to control one lamp, fan, and three-pin outlet socket independently by single-way switches placed on a switch board:** The circuit is shown in Figure 18.8. A three-pin outlet socket, a fan regulator, and three switches are fixed on a switch board symmetrically as shown in Figure 18.8. The phase wire is given to all the appliances (i.e., lamp, fan, and three-pin outlet socket) through switches, whereas neutral wire is connected directly. An earth wire is connected to the earth terminal (large terminal) of three-pin outlet socket. In the fan circuit, a fan regulator is connected in series to regulate the fan speed.

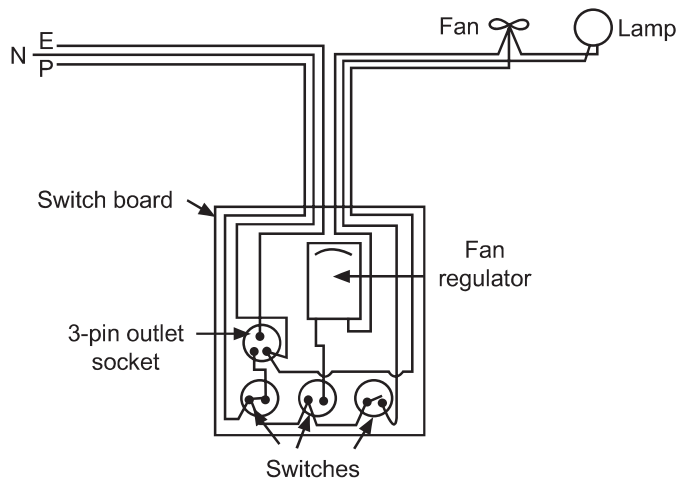


Fig. 18.8 Switch board to control one lamp, one tube and on fan with regulator

4. **Circuit to control three-phase induction motor:** The circuit diagram to control a three-phase induction motor is shown in Figure 18.9. Only three wires are run from

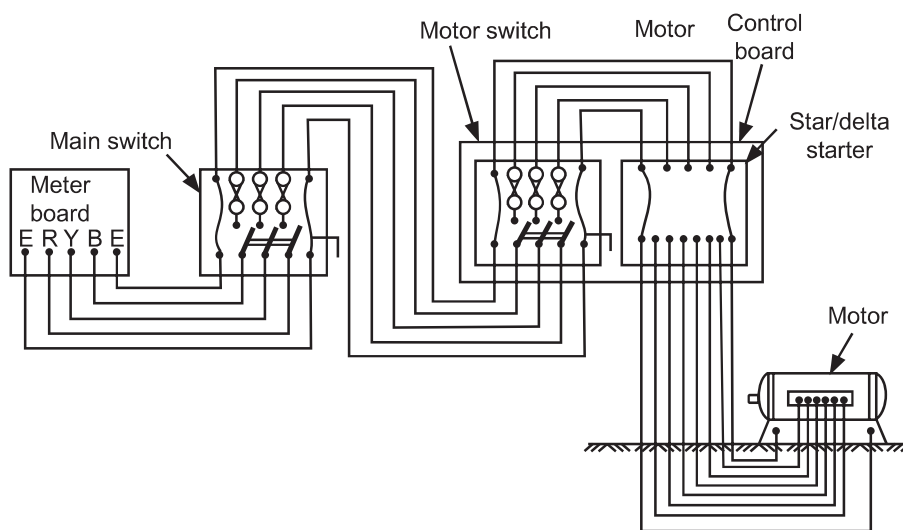


Fig. 18.9 Electric wiring circuit to control 3-phase induction motor through a star-delta starter